

Occupancy Sensor API Specification

Required Endpoints

GET `/ {building} / {floor} / sensors`

Returns all sensor names on a given floor in a given building.

Response — 200 OK

```
{
  "building": "Al-Midra Tower",
  "floor": 3,
  "sensors": [
    "Sensor1",
    "Sensor2",
    "Sensor3"
  ]
}
```

Error responses

HTTP status	error value	Meaning
404	"floor_not_found"	No floor with that number exists in the specified building
404	"building_not_found"	No building with that name exists

```
{ "error": "floor_not_found", "message": "Floor 3 not found in Al-Midra Tower." }
```

GET `/ {building} / {floor} / {sensor_name}`

Returns the layout and occupancy data for a specific sensor over a given time window.

Query parameters

Parameter	Type	Required	Description
start	string (ISO 8601)	Yes	Start of the time window, e.g. 2025-01-01T08:00:00Z
end	string (ISO 8601)	Yes	End of the time window, e.g. 2025-01-31T18:00:00Z

Response — 200 OK

```
{
  "sensor_name": "Sensor1",
  "bounding_box": {
    "x1": 400,
    "y1": 180,
    "x2": 520,
    "y2": 300
  },
  "seats": [
    {
      "seat_id": "Sensor1-001",
      "x": 412,
      "y": 208,
      "assigned": true,
      "organization_code": "1231322",
      "organization_name": "Finance",
      "business_line": "Upstream",
      "admin_area": "Dhahran",
      "department": "Reservoir Engineering",
      "division": "Well Performance",
      "occupancy_rate": 0.74
    },
    {
      "seat_id": "Sensor1-002",
      "x": 445,
      "y": 208,
      "assigned": false,
      "organization_code": null,
      "organization_name": null,
      "business_line": null,
      "admin_area": null,
      "department": null,
      "division": null,
      "occupancy_rate": 0.12
    }
  ]
}
```

Response fields

Top-level

Field	Type	Description
sensor_name	string	Echo of the requested sensor name
bounding_box	object	Pixel coordinates of the top-left (x1, y1) and bottom-right (x2, y2) corners of the area covered by this sensor
seats	array	All seats within this sensor's coverage area.

bounding_box object

Field	Type	Description
x1	integer	Pixel X of the top-left corner
y1	integer	Pixel Y of the top-left corner
x2	integer	Pixel X of the bottom-right corner
y2	integer	Pixel Y of the bottom-right corner

Seat object

Field	Type	Description
seat_id	string	Stable unique identifier for the seat
x	integer	Pixel X of the seat's centre point, where top-left corner of the sensor's coverage area is (0, 0)
y	integer	Pixel Y of the seat's centre point, where top-left corner of the sensor's coverage area is (0, 0)
assigned	boolean	true if this seat is assigned to an organization, false otherwise
organization_code	string null	Code for the occupying organization. null if assigned is false
organization_name	string null	Full name of the occupying organization. null if assigned is false
business_line	string null	Business line of the occupying organization. null if assigned is false
admin_area	string null	Admin area. null if assigned is false
department	string null	Department name. null if assigned is false
division	string null	Division name. null if assigned is false

occupancy_rate	float	Fraction of time this seat was occupied during the requested window (0.0–1.0). Returned regardless of assignment status.
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Error responses

HTTP status	error value	Meaning
404	"sensor_not_found"	No sensor with that name exists on the specified building and floor

```
{ "error": "sensor_not_found", "message": "No sensor named 'Sensor1' found on floor 3 of Al-Midra Tower." }
```